

## A Membership inference (user-level)

### Deployment A

Target model  
risk scores  $\tilde{p}_u$



### Deployment B

Attack model  
confidence / entropy  
+ shadow models



### Metrics

MIA AUC  
TPR@1%FPR

### Decision

$\tilde{m} \in \{0, 1\}$   
in training or not?

## B Text Reconstruction Stress Test

Dense embeddings  
MiniLM baseline



Generative inversion  
Vec2Text guidance

Hashed + DP (ours)  
 $\tilde{\varphi}$  + concepts



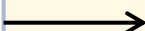
HIGH overlap  
+PII risk

LOW fidelity  
(reverse-hash)

## C Cross-deployment linkage



Representation  $v$   
(deploy A: salt  $\sigma_1, \epsilon_1$ )



Representation  $V'$   
(deploy B: salt  $\sigma_2, \epsilon_2$ )



Pairwise matcher  
cosine/JS divergence+  
threshold sweep



Match / non-match  
same user or not?



### Linkage Resistance

Salt rotation  $\sigma_1 \neq \sigma_2 + \epsilon - LDP \rightarrow$  lower linkage AUC  
Metric: linkage AUC (lower is better)